The Four-Gamete Condition for Perfect Phylogenies

Theoretical part of the project *Mutations*

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Abstract

We give a complete and self-contained proof of the following claim. Let A be an $n \times m$ matrix of DNA sequences in which every column contains at most two distinct letters. Then A admits a perfect phylogeny if and only if it satisfies the *four-gamete test*: for every pair of columns, the rows restricted to those two columns exhibit at most three of the four possible configurations. The proof reduces the claim to the containment theorem for rooted binary matrices proved in the lecture, by analysing what Step 0 of the perfect-phylogeny algorithm (choosing a reference taxon and re-encoding every column relative to it) does to a pair of columns. For completeness, the containment theorem itself is proved in the appendix.

1 Setting and definitions

Throughout, $\Sigma = \{\mathsf{A}, \mathsf{C}, \mathsf{G}, \mathsf{T}\}$ is the DNA alphabet, $S = \{1, \ldots, n\}$ with $n \geq 1$ is a set of *taxa*, and $A \in \Sigma^{n \times m}$ is a matrix whose i-th row A_i is the sequence of taxon i and whose j-th column is denoted A_j. We write A_{ij} for the letter of taxon i at position j. Positions are also called *characters*.

Definition 1.1 (Binary column)**.** Column j is *binary* if $\Sigma_j \coloneqq \{A_{ij} : i \in S\}$ has at most two elements. The matrix A is binary if all of its columns are.

All graphs in this note are finite trees. A *leaf* is a vertex of degree at most one. For two vertices u, v of a tree T we write $\pi(u, v)$ for the set of edges on the unique path between them.

Definition 1.2 (Perfect phylogeny)**.** A *perfect phylogeny* for A is a pair (T, σ) where T is a tree and $\sigma \colon V(T) \to \Sigma^m$ assigns a sequence to every vertex, such that

(P1) the taxa are placed at distinct leaves of T, every leaf of T holds a taxon, and $\sigma(v) = A_i$ whenever leaf v holds taxon i;

(P2) for every character j, at most one edge $\{u, v\} \in E(T)$ satisfies $\sigma(u)_j \neq \sigma(v)_j$.

An edge on which character j changes is called a *mutation edge* of j. Nothing in this definition, or in Lemma 1.3 below, depends on the alphabet: both apply verbatim to matrices over any finite alphabet, in particular over $\{0, 1\}$.

Condition (P2) says that every character mutates at most once in the whole tree; a character that is constant on S need not mutate at all. Note that T is unrooted and that no assumption is made on the sequences of internal vertices beyond (P2). The following lemma shows that internal vertices cannot carry "invented" letters.

Lemma 1.3 (No invented states)**.** *Let (T, σ) be a perfect phylogeny for A. Then $\sigma(v)_j \in \Sigma_j$ for every vertex v and every character j.*

Proof. Fix j. By (P2), removing the mutation edge of j (if there is one) from T leaves at most two connected components, and $\sigma(\cdot)_j$ is constant on each component, because no edge inside a component is a mutation edge of j. It therefore suffices to show that every component contains a leaf of T. If no edge is removed this is trivial. Otherwise let $e = \{u, v\}$ be removed and consider the component K containing u. If $K = \{u\}$ then u has degree one in T, so u is a leaf of T. If K has at least two vertices then K, being a tree, has at least two leaves, and at least one of them is different from u; such a leaf has the same degree in K as in T, hence is a leaf of T. By (P1) every leaf of T holds a taxon, so the constant value of $\sigma(\cdot)_j$ on each component is a letter that occurs in column j. □

Definition 1.4 (Four-gamete test)**.** For two columns $j \neq l$ of A, the *configuration* of row i on (j, l) is the pair (A_{ij}, A_{il}). The matrix A *satisfies the four-gamete test* if for every pair of columns $j \neq l$ the set of configurations $\{(A_{ij}, A_{il}) : i \in S\}$ has at most three elements.

For a binary matrix each column has at most two letters, so a pair of columns has at most four possible configurations, and the test says that not all four occur. We can now state the claim.

Theorem 1.5 (Main claim)**.** *A binary DNA matrix A has a perfect phylogeny if and only if it satisfies the four-gamete test.*

2 The rooted formulation from the lecture

The perfect-phylogeny algorithm presented in the lecture works with 0/1 matrices in which a chosen taxon has all zeros. We recall that setting and the theorem proved there.

Definition 2.1 (Step 0)**.** Let A be binary and fix a reference taxon $r \in S$. For every column j define the map $\varphi_j \colon \Sigma_j \to \{0, 1\}$ by $\varphi_j(A_{rj}) = 0$ and $\varphi_j(x) = 1$ for the other letter $x \in \Sigma_j$ (if any). The *rooted matrix* $C = C(A, r) \in \{0, 1\}^{n \times m}$ is $C_{ij} = \varphi_j(A_{ij})$. For every column j let

$$O_j \coloneqq \{\, i \in S : C_{ij} = 1 \,\}$$

be the set of taxa carrying the derived state of character j. Column j is *trivial* if $O_j = \emptyset$, i.e. if A_j is constant.

Since Σ_j has at most two letters, each φ_j is injective, and row r of C is the all-zero vector. More generally, for any $C \in \{0, 1\}^{n \times m}$ we write $O_j \coloneqq \{i \in S : C_{ij} = 1\}$ and call column j trivial if $O_j = \emptyset$; Definition 2.2 and the results below use this notation for arbitrary 0/1 matrices.

Definition 2.2 (Rooted perfect phylogeny)**.** Let $C \in \{0, 1\}^{n \times m}$. A *rooted perfect phylogeny* for C is a tree T with a distinguished vertex ρ (the *root*), together with an assignment $j \mapsto e_j \in E(T)$ of every non-trivial column to an edge, such that

(R1) the taxa are placed at distinct leaves of T, every leaf of T other than ρ holds a taxon, and ρ holds no taxon;

(R2) for every taxon i (placed at leaf v_i), $\{\, j : e_j \in \pi(\rho, v_i) \,\} = \{\, j : C_{ij} = 1 \,\}$.

Several columns may be assigned to the same edge. Condition (R2) says that reading the labels on the path from the root to taxon i recovers exactly the ones in row i: each character mutates from 0 to 1 exactly once, on the edge e_j, and O_j is the set of taxa below e_j. In the lecture the root *is* the all-zero taxon r; here the root ρ holds no taxon and r sits at a leaf. The two formulations are equivalent (attach v_r to ρ by a single unlabelled edge, or contract that edge), exactly as in the proof of Lemma 3.2; we use the taxon-free root because it makes the inductive construction in Appendix A cleaner.

Theorem 2.3 (Containment theorem; the lecture's theorem in the formulation of Definition 2.2)**.** *A matrix $C \in \{0,1\}^{n\times m}$ has a rooted perfect phylogeny if and only if for every two columns $j \neq l$*

$$O_j \cap O_l = \emptyset \quad or \quad O_j \subseteq O_l \quad or \quad O_l \subseteq O_j. \tag{$\star$}$$

A family of sets satisfying ($\star$) is called *laminar*. For completeness a full proof of Theorem 2.3 is given in Appendix A; the body of this note uses it as a black box.

3 Proof of the main claim

Proof in brief. Fix a taxon r and apply Step 0: every column is re-encoded so that r's letter is 0 and the other letter is 1. Since each column has at most two letters, this re-encoding is a bijection on the letters of every column, so it changes neither the set of row configurations of any pair of columns (hence not the four-gamete test) nor the existence of a perfect phylogeny. In the resulting matrix row r is all zeros, so every pair of columns already shows the configuration 00; "at most three configurations" therefore means "not all of 11, 10, 01". But 11 appears iff $O_j \cap O_l \neq \emptyset$, 10 appears iff $O_j \not\subseteq O_l$, and 01 appears iff $O_l \not\subseteq O_j$; so the four-gamete test is exactly the condition that every two sets O_j, O_l are disjoint or nested, which by the containment theorem of the lecture is equivalent to the existence of a rooted perfect phylogeny. The rest of this section makes each of these steps precise: that Step 0 preserves perfect phylogenies (Lemma 3.1), that rooted and unrooted perfect phylogenies of the re-encoded matrix are the same thing (Lemma 3.2), the translation of configurations into set relations (Lemma 3.3) and the invariance of the test (Lemma 3.4).

The proof is a chain of equivalences, established by the lemmas below and assembled in Section 3.4:

$$\begin{aligned} A \text{ has a PP} &\overset{\text{L.3.1}}{\iff} C \text{ has a PP} \overset{\text{L.3.2}}{\iff} C \text{ has a rooted PP} \overset{\text{Thm.2.3}}{\iff} (\star) \text{ holds for all pairs} \\ &\overset{\text{L.3.3}}{\iff} C \text{ passes the four-gamete test} \overset{\text{L.3.4}}{\iff} A \text{ passes the four-gamete test.} \end{aligned}$$

3.1 Step 0 preserves the existence of a perfect phylogeny

Lemma 3.1. *Let A be binary, $r \in S$, and $C = C(A, r)$. Then A has a perfect phylogeny if and only if C has one (in the sense of Definition 1.2, with alphabet $\{0,1\}$).*

Proof. ($\Rightarrow$) Let (T, σ) be a perfect phylogeny for A. By Lemma 1.3, $\sigma(v)_j \in \Sigma_j$ for all v, j, so $\sigma'(v)_j := \varphi_j(\sigma(v)_j)$ is well defined. Condition (P1) holds for (T, σ') and C because $\sigma'(v_i)_j = \varphi_j(A_{ij}) = C_{ij}$. Since φ_j is injective on Σ_j, for every edge $\{u, v\}$ we have $\sigma'(u)_j \neq \sigma'(v)_j$ if and only if $\sigma(u)_j \neq \sigma(v)_j$; hence the mutation edges of every character are the same as before and (P2) holds.

($\Leftarrow$) Let (T, τ) be a perfect phylogeny for C. By Lemma 1.3 applied to C, $\tau(v)_j \in \{C_{ij} : i \in S\} = \varphi_j(\Sigma_j)$, so $\sigma(v)_j := \varphi_j^{-1}(\tau(v)_j)$ is well defined, and the same argument (with the injective map φ_j^{-1}) shows that (T, σ) is a perfect phylogeny for A. □

3.2 Unrooted and rooted perfect phylogenies of C coincide

Lemma 3.2. *Let $C \in \{0,1\}^{n\times m}$ have an all-zero row r. Then C has a perfect phylogeny (Definition 1.2) if and only if it has a rooted perfect phylogeny (Definition 2.2).*

Proof. ($\Rightarrow$) Let (T, τ) be a perfect phylogeny for C and let v_r be the leaf holding r. If T is the single vertex v_r then $n = 1$, all columns are trivial, and the tree consisting of a new root ρ joined to v_r by one edge is a rooted perfect phylogeny. Otherwise let w be the neighbour of v_r; subdivide the edge $\{v_r, w\}$ by a new vertex ρ and set $\tau(\rho) := \tau(v_r) = C_r = 0$. Subdividing an

edge does not change the leaves of T. Since $\tau(\rho) = \tau(v_r)$, the new edge $\{v_r, \rho\}$ is a mutation edge of no character, and $\{\rho, w\}$ is a mutation edge of j exactly when $\{v_r, w\}$ was; so the number of mutation edges of every character is unchanged, (T, τ) is still a perfect phylogeny, and ρ holds no taxon, giving (R1).

Let j be a non-trivial column. Some taxon i has $C_{ij} = 1 \neq 0 = \tau(\rho)_j$, so at least one edge on $\pi(\rho, v_i)$ is a mutation edge of j; by (P2) there is exactly one mutation edge of j in T, call it e_j. Now let i be any taxon. Walking along $\pi(\rho, v_i)$ from $\tau(\rho)_j = 0$, the value of character j flips exactly when the walk crosses e_j, and e_j is crossed at most once because paths in a tree are simple. Therefore $C_{ij} = \tau(v_i)_j = 1$ if and only if $e_j \in \pi(\rho, v_i)$, which is (R2).

($\Leftarrow$) Let $(T, \rho, (e_j)_j)$ be a rooted perfect phylogeny for C. Define $\tau(v)_j \coloneqq 1$ if j is non-trivial and $e_j \in \pi(\rho, v)$, and $\tau(v)_j \coloneqq 0$ otherwise. By (R2), $\tau(v_i) = C_i$ for every taxon i, and by (R1) the taxa sit at distinct leaves.

We first verify (P2) for (T, τ). Let $\{u, v\}$ be an edge with u closer to ρ than v; then $\pi(\rho, v) = \pi(\rho, u) \cup \{\{u, v\}\}$, so $\tau(u)_j \neq \tau(v)_j$ if and only if $\{u, v\} = e_j$. Thus every non-trivial character has exactly one mutation edge, namely e_j, and trivial characters have none.

The pair (T, τ) may fail (P1) only because some leaf holds no taxon (by (R1) this can only be ρ). We repair this by pruning: as long as T has a leaf holding no taxon, delete that leaf together with its incident edge. Each deletion removes an edge, so the number of mutation edges of any character cannot increase and (P2) is preserved; it does not change τ on the remaining vertices, so the taxa keep their rows; and it does not change the degree of any remaining vertex except the deleted leaf's neighbour w. If w holds a taxon then w was a leaf whose only neighbour was the deleted leaf, so T was a single edge and w becomes an isolated vertex, which is still a leaf. Hence taxa remain at distinct leaves throughout. The process terminates because T is finite, and when it stops every leaf holds a taxon, so (P1) holds. (Note that a non-taxon vertex of degree two, which ρ may well be, is not a leaf and is allowed by Definition 1.2.) □

3.3 What Step 0 does to a pair of columns

The heart of the argument is the following elementary translation between row configurations and set relations. Recall that row r of C is all zeros.

Lemma 3.3 (Configurations versus sets)**.** *Let $C \in \{0,1\}^{n \times m}$ have an all-zero row r, and let $j \neq l$ be two columns. Write $\langle ab \rangle$ for the statement "some row i has $(C_{ij}, C_{il}) = (a, b)$". Then*

(i) $\langle 00 \rangle$ *always holds;*

(ii) $\langle 11 \rangle$ *holds if and only if $O_j \cap O_l \neq \emptyset$;*

(iii) $\langle 10 \rangle$ *holds if and only if $O_j \not\subseteq O_l$;*

(iv) $\langle 01 \rangle$ *holds if and only if $O_l \not\subseteq O_j$.*

Consequently, the pair (j, l) exhibits at most three configurations if and only if $O_j \cap O_l = \emptyset$ or $O_j \subseteq O_l$ or $O_l \subseteq O_j$; that is, if and only if $(\star)$ holds for (j, l).

Proof. (i) Row r has configuration $(0, 0)$. (ii) A row i with $(C_{ij}, C_{il}) = (1, 1)$ is exactly an element of $O_j \cap O_l$. (iii) A row i with $(1, 0)$ is exactly a taxon in $O_j \setminus O_l$, and $O_j \setminus O_l \neq \emptyset$ if and only if $O_j \not\subseteq O_l$. (iv) is (iii) with the roles of j and l exchanged.

For the consequence: by (i) the configuration $(0, 0)$ is always present, so "at most three configurations" is equivalent to "not all of $\langle 11 \rangle, \langle 10 \rangle, \langle 01 \rangle$ hold". By (ii)–(iv) this is equivalent to

$$O_j \cap O_l = \emptyset \quad \text{or} \quad O_j \subseteq O_l \quad \text{or} \quad O_l \subseteq O_j,$$

which is $(\star)$. □

Lemma 3.4 (Step 0 preserves the four-gamete test)**.** *Let A be binary, $r \in S$, and $C = C(A, r)$. For every pair of columns $j \neq l$, the number of distinct configurations of (j, l) in A equals the number of distinct configurations of (j, l) in C. In particular A satisfies the four-gamete test if and only if C does.*

Proof. The map $\Phi\colon \Sigma_j \times \Sigma_l \to \{0,1\}^2$, $\Phi(a, b) = (\varphi_j(a), \varphi_l(b))$, is injective because φ_j and φ_l are injective on Σ_j and Σ_l respectively. The configuration of row i in C is Φ applied to its configuration in A, so Φ restricts to a bijection between the two sets of configurations. □

3.4 Conclusion

Proof of Theorem 1.5. Fix any taxon r and let $C = C(A, r)$. Then

$$
\begin{aligned}
A \text{ has a perfect phylogeny} &\iff C \text{ has a perfect phylogeny} && \text{(Lemma 3.1)}\\
&\iff C \text{ has a rooted perfect phylogeny} && \text{(Lemma 3.2)}\\
&\iff (\star) \text{ holds for every pair } j \neq l && \text{(Theorem 2.3)}\\
&\iff C \text{ satisfies the four-gamete test} && \text{(Lemma 3.3)}\\
&\iff A \text{ satisfies the four-gamete test.} && \text{(Lemma 3.4)}
\end{aligned}
$$
□

Remark 3.5 (Why it is called the four-gamete test). For two binary characters, the four configurations $(0,0), (0,1), (1,0), (1,1)$ are the four possible "gametes" (haplotypes) on two sites. In a perfect phylogeny each character mutates once, so the two mutation edges cut the tree into at most three regions, and all taxa inside one region share the same configuration; hence at most three of the four gametes can be observed. Theorem 1.5 says that this obvious necessary condition is also sufficient. The test appears in population genetics as a test for recombination [6], and the underlying compatibility criterion for binary characters goes back to [3]; see [5, Ch. 17] for the algorithmic treatment.

Example 3.6. Consider the five sequences below (the example from the lecture, written over Σ), with $r = E$.

	1	2	3	4
A	G	C	T	A
B	A	C	T	A
C	A	C	T	G
D	A	T	C	G
E	A	C	C	G

$\xrightarrow{\text{Step 0}}$

	1	2	3	4
A	1	0	1	1
B	0	0	1	1
C	0	0	1	0
D	0	1	0	0
E	0	0	0	0

Here $O_1 = \{A\}$, $O_2 = \{D\}$, $O_3 = \{A, B, C\}$, $O_4 = \{A, B\}$, a laminar family, and indeed no pair of columns shows four configurations (for instance columns $3, 4$ show only $11, 10, 00$). Figure 1 shows a perfect phylogeny. Changing D's letter at position 4 to A would give $O_4 = \{A, B, D\}$, which neither contains, nor is contained in, nor is disjoint from O_3; the pair $(3, 4)$ would then show all four configurations (A:11, C:10, D:01, E:00) and no perfect phylogeny would exist.

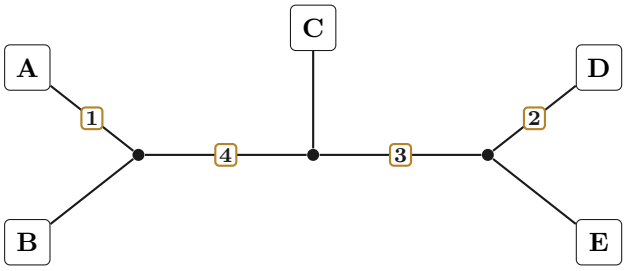


Figure 1: An (unrooted) perfect phylogeny for the matrix of Example 3.6. Each character labels the single edge on which it mutates; the three unlabelled pendant edges carry no mutation. Rooting the tree at E recovers the rooted phylogeny of the lecture.

Remark 3.7 (Scope of the binary assumption). The theorem, and the $O(nm)$ algorithm of [4], rely on every column having at most two states. For characters with three or more states the perfect phylogeny problem is NP-complete [2, 7], although it becomes polynomial when the number of states is fixed [1]. In DNA data the binary assumption holds for sites at which only two nucleotides are observed, which is the typical situation for closely related sequences.

A Proof of the containment theorem

We prove Theorem 2.3: a 0/1 matrix C has a rooted perfect phylogeny if and only if the family $\{O_j\}_j$ is laminar, i.e. $(\star)$ holds for all $j \neq l$. Trivial (all-zero) columns can be ignored on both sides: they are not assigned to edges and contribute nothing to either side of (R2), so a rooted perfect phylogeny for C is one for C with its trivial columns deleted and conversely; and the empty set satisfies $(\star)$ together with every set. We may and do assume that all columns are non-trivial. We use the following standard fact.

Lemma A.1. *Let T be a tree with root ρ and let $e = \{u, v\} \in E(T)$ with u closer to ρ than v. Write* below(e) *for the vertex set of the component of $T - e$ containing v. Then for every vertex x, $e \in \pi(\rho, x)$ if and only if $x \in$* below(e)*. Moreover, if e, e' are two edges and e lies on the path from ρ to the endpoint of e' farther from ρ, then* below$(e') \subseteq$ below(e).

Proof. The path from ρ to x uses e exactly when ρ and x lie in different components of $T - e$, and $\rho \notin$ below(e). For the second statement, if $e \in \pi(\rho, v')$ for the far endpoint v' of e', then for any $x \in$ below(e') the path $\pi(\rho, x)$ passes through e' and hence through v', so it contains $\pi(\rho, v') \ni e$; thus $x \in$ below(e). □

In a rooted perfect phylogeny, (R2) together with Lemma A.1 says precisely that

$$O_j = \{\, i \in S : v_i \in \text{below}(e_j) \,\} \qquad \text{for every column } j. \tag{1}$$

Necessity ($\Rightarrow$)

Let $(T, \rho, (e_j)_j)$ be a rooted perfect phylogeny for C and let $j \neq l$ with $O_j \cap O_l \neq \emptyset$; we show that one of O_j, O_l contains the other. Pick $i \in O_j \cap O_l$. By (R2) both e_j and e_l lie on the path $\pi(\rho, v_i)$. If $e_j = e_l$ then $O_j = O_l$ by (1). Otherwise one of them, say e_j, is encountered first when walking from ρ to v_i, so e_j lies on the path from ρ to the endpoints of e_l, and Lemma A.1 gives below$(e_l) \subseteq$ below(e_j), hence $O_l \subseteq O_j$ by (1). The other case gives $O_j \subseteq O_l$.

Sufficiency ($\Leftarrow$)

We prove by strong induction on the number m of columns that every matrix $C \in \{0,1\}^{n \times m}$ (on any non-empty taxon set S) with non-trivial columns whose family $\{O_j\}$ is laminar has a rooted perfect phylogeny.

Base, $m = 0$. Take T to be the star with centre ρ and one leaf v_i for each taxon $i \in S$ (for $|S| = 1$ this is a single edge). There are no columns, so (R2) holds vacuously, and (R1) holds.

Step, $m \geq 1$. Choose a column j with $|O_j|$ maximum. For every other column l, laminarity gives $O_l \cap O_j = \emptyset$, or $O_l \subseteq O_j$, or $O_j \subseteq O_l$; in the last case maximality forces $O_l = O_j$, so in all cases

$$O_l \subseteq O_j \quad \text{or} \quad O_l \cap O_j = \emptyset.$$

Since every O_l is non-empty, exactly one of the two alternatives holds for each $l \neq j$. Let

$$P \coloneqq O_j, \qquad Q \coloneqq S \setminus P, \qquad L_1 \coloneqq \{\, l \neq j : O_l \subseteq P \,\}, \qquad L_2 \coloneqq \{\, l \neq j : O_l \cap P = \emptyset \,\}.$$

Then $P \neq \emptyset$, $\{L_1, L_2\}$ partitions the columns other than j, and:

- for $l \in L_1$ and $i \in Q$: $C_{il} = 0$ (because $O_l \subseteq P$);
- for $l \in L_2$ and $i \in P$: $C_{il} = 0$ (because $O_l \cap P = \emptyset$).

Let C_1 be the submatrix of C with rows P and columns L_1, and C_2 the submatrix with rows Q and columns L_2 (if $Q = \emptyset$, then every O_l, $l \neq j$, is contained in $P = S$, so $L_2 = \emptyset$; we then take C_2 to be the empty matrix on an empty taxon set and let T_2 be the single vertex ρ_2). In C_1 the set of taxa with a 1 in column $l \in L_1$ is still O_l (all of O_l lies in P), and in C_2 it is still O_l for $l \in L_2$; hence both families are laminar and non-trivial, and both matrices have fewer than m columns. By the induction hypothesis C_1 has a rooted perfect phylogeny $(T_1, \rho_1, (e_l)_{l \in L_1})$ and C_2 has one, $(T_2, \rho_2, (e_l)_{l \in L_2})$.

Form T from the disjoint union of T_1 and T_2 by adding the edge $e_j \coloneqq \{\rho_2, \rho_1\}$, and declare $\rho \coloneqq \rho_2$ to be the root (Figure 2). We check the two conditions.

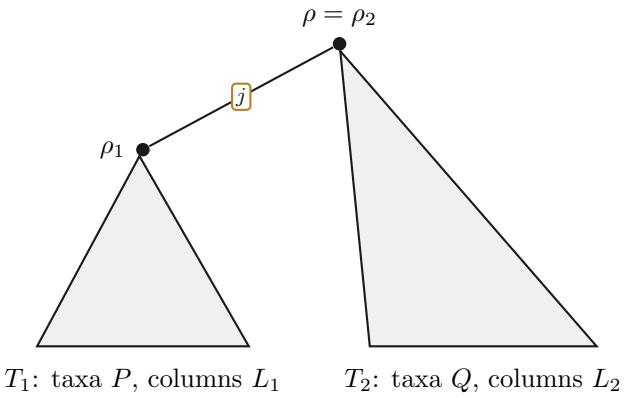


Figure 2: The inductive construction: the two recursively built phylogenies are joined by a new edge carrying character j, and the root of T_2 becomes the root of the whole tree.

(R1): the taxa of P sit at distinct leaves of T_1 and those of Q at distinct leaves of T_2, and these remain leaves of T because the only new edge is incident to ρ_1 and ρ_2, which hold no taxa. Since $P \neq \emptyset$, T_1 has a taxon at a leaf other than ρ_1, so ρ_1 has degree at least one in T_1 and at least two in T; hence ρ_1 is not a leaf of T, and every leaf of T other than possibly $\rho = \rho_2$ holds a taxon.

(R2): let $i \in P$. The path from ρ_2 to v_i consists of the edge e_j followed by the path from ρ_1 to v_i inside T_1. The columns assigned to its edges are therefore $\{j\} \cup \{l \in L_1 : C_{il} = 1\}$. Since $C_{ij} = 1$ (as $i \in P = O_j$) and $C_{il} = 0$ for all $l \in L_2$, this set equals $\{l : C_{il} = 1\}$. Now let $i \in Q$. The path from ρ_2 to v_i stays inside T_2, so the columns on it are $\{l \in L_2 : C_{il} = 1\}$; since $C_{ij} = 0$ and $C_{il} = 0$ for all $l \in L_1$, this again equals $\{l : C_{il} = 1\}$.

Hence $(T, \rho, (e_l)_l)$ is a rooted perfect phylogeny for C, completing the induction. □

Remark A.2. The construction is exactly the recursive algorithm of the lecture: choose a maximal character, split the taxa into O_j and its complement, recurse, and join. The proof of necessity is also what the algorithm uses to *detect* that no perfect phylogeny exists: if some O_l is neither contained in nor disjoint from the chosen O_j, the pair (j, l) violates $(\star)$, and by Lemma 3.3 the four rows witnessing the configurations $11, 10, 01, 00$ form an explicit four-gamete certificate.

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